<u>Habib University</u>

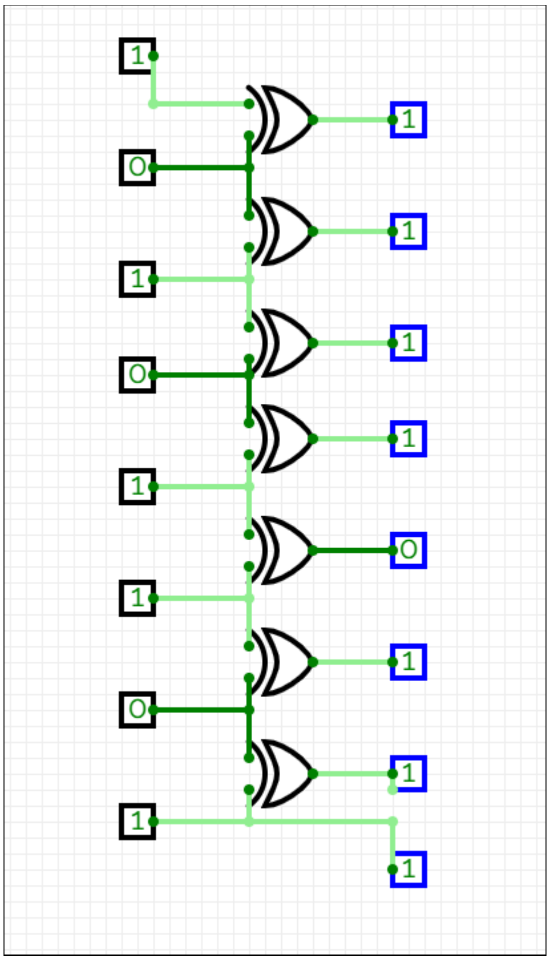
<u>Digital Logic and Design</u>

<u>EE/CE-172/222</u>

Assignment 2 Fall 2025 Solution Manual

Question 1:

(a)



(b)

Binary counting can cause many bits to change simultaneously. Gray encoding guarantees that consecutive values differ in only one bit. During transitions, multiple simultaneous changes can cause errors or unstable signals. If Faaz's steps weren't converted to Gray code, these errors could be misread as wrong steps and Faaz may fall before creating the longest assignment.

Question: 02(a) 2pts

$\rightarrow (x+y+z')' + (x'+y'+z)' + (w+x+y+z')' + (w'+x'+y+z)'$

$\rightarrow x'y'z + xyz' + w'x'y'z + wxy'z'$

x	y	z	w	x+y+z’	x’+y’+z	w+x+y+z’	w’+x’+y+z	F	x’y’z	xyz’	x’y’zw’	xy’z’w	F`
0	0	0	0	1	1	1	1	**1**	0	0	0	0	**0**
0	0	0	1	1	1	1	1	**1**	0	0	0	0	**0**
0	0	1	0	0	1	0	1	**0**	1	0	1	0	**1**
0	0	1	1	0	1	1	1	**0**	1	0	1	0	**1**
0	1	0	0	1	1	1	1	**1**	0	0	0	0	**0**
0	1	0	1	1	1	1	1	**1**	0	0	0	0	**0**
0	1	1	0	1	1	1	1	**1**	0	0	0	0	**0**
0	1	1	1	1	1	1	1	**1**	0	0	0	0	**0**
1	0	0	0	1	1	1	1	**1**	0	0	0	0	**0**
1	0	0	1	1	1	1	0	**0**	0	0	0	1	**1**
1	0	1	0	1	1	1	1	**1**	0	0	0	0	**0**
1	0	1	1	1	1	1	1	**1**	0	0	0	0	**0**
1	1	0	0	1	0	1	1	**0**	0	1	0	0	**1**

1	1	0	1	1	0	1	1	**0**	0	1	0	0	**1**
1	1	1	0	1	1	1	1	**1**	0	0	0	0	**0**
1	1	1	1	1	1	1	1	**1**	0	0	0	0	**0**

(b)

$(a+b')(c+d')(a'+c)(b+d)(a'+b'+c'+d')$ 2pts

$(ab'+ac'+ad'+a'b'+b'+b'c'+b'd')(c+d')(a'b+a'd+bc+cd)$

$(b'[a+1+c'+d']+ac'+ad'+a'b')(c+d')(a'b+a'd+bc+cd)$

$(b'+ac'+ad'+a'b')(a'bc+a'cd+bc+cd+a'bd'+bcd')$

$(b'+ac'+ad')(bc(a'+1+d')+cd(a'+1)+a'bd')$

$(b'+ac'+ad')(bc+cd+a'bd')$

$\boxed{b'cd + abcd'}$ 7 literal   2 terms

(c)   4 pts

A → 1 1 0 1 0 1 1 1 0 0 1 1
B → 1 0 1 1 1 1 0 0 1 0 1 0

A ⊗ B → 0 1 1 0 1 0 1 1 1 0 0 1

A ∨ B → 1 1 1 1 1 1 1 1 1 0 1 1

A' → 0 0 1 0 1 0 0 0 1 1 0 0

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Question: 03 (a) 2pts

$F_1 = ac + ad + b'c + b'd$

$F_{1(min)} = abcd + ab'cd + abcd' + ab'cd' + ab'c'd + abc'd + a'b'cd + a'b'cd' + a'b'c'd$

So, $F_1 = \sum(1,2,3,9,10,11,13,14,15)$

$F_2 = (ab' + cd)(a' + d)$

$F_2 = ab'd + a'cd + cd$

$$\begin{array}{ccccc} 1011 & 1001 & 0111 & 0011 & 1111 \end{array}$$

$F_{2(min)} = ab'cd + ab'c'd + a'bcd + a'b'cd + abcd$

$F_2 = \sum(3,7,9,11,15)$

Lets compute $E = F_1 + F_2 = ac + ad + b'c + b'd + ab'd + a'cd + cd$

$E = ac + ad + b'c + b'd + cd$

$E_{(min)} = abcd + ab'cd + abc'd + ab'c'd + abcd' + ab'cd' + a'bcd + a'b'cd + a'b'cd' + a'b'c'd$

$E_{(min)} = \sum(1,2,3,7,9,10,11,13,14,15)$

Hence proved that E contains sum of minterms of F_1 and F_2

(b) 2pts

Lets compute $G = F_1 . F_2 =$

$G = (ac + ad + b'c + b'd) + \cancel{cd} . (ab'd + a'cd + cd)$

$G = ab'cd + acd + ab'd + acd + ab'cd + a'b'cd + b'cd + b'ad + a'b'cd + b'cd$

$G = acd + \cancel{ab'd}\ b'cd + a'b'cd + ab'd$

$$\begin{array}{cccc} 1111 & 1011 & 0011 & 1001 \end{array}$$

$G_{(min)} = abcd + ab'cd + a'b'cd + ab'c'd$

$G = \sum(3,9,11,15)$ which are the common minterms of F_1 and F_2

Proved

a	**b**	**c**	**d**	**F1**	**F2**	**E=cd+b’d+ac+ad+b’c**	**G=F1·F2**
0	0	0	0	0	0	**0**	**0**
0	0	0	1	**1**	0	**1**	**0**
0	0	1	0	**1**	0	**1**	**0**
0	0	1	1	**1**	**1**	**1**	**1**
0	1	0	0	0	0	**0**	**0**
0	1	0	1	0	0	**0**	**0**
0	1	1	0	0	0	**0**	**0**
0	1	1	1	0	**1**	**1**	**0**
1	0	0	0	0	0	**0**	**0**
1	0	0	1	**1**	**1**	**1**	**1**
1	0	1	0	**1**	0	**1**	**0**
1	0	1	1	**1**	**1**	**1**	**1**
1	1	0	0	0	0	**0**	**0**
1	1	0	1	**1**	0	**1**	**0**
1	1	1	0	**1**	0	**1**	**0**
1	1	1	1	**1**	**1**	**1**	**1**

Question: 04 (a) 2pts

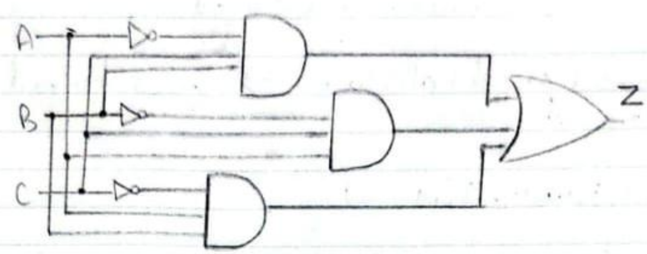
A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

(b) 2pts

Minterms: $Z = A'BC + AB'C + ABC'$, $Z = \Sigma(3,5,6)$

Maxterms: $Z = (A+B+C)(A+B+C')(A+B'+C)(A'+B+C)(A'+B'+C')$

(c) 2pts



Question: 05 3pts

$\overline{F}^{*} = [(ab+cd')' + (a'b'c+d')' + (a+b'+c')' + (ab'+cd)' + (b+c+d')']$

Considering all terms individually:

$T_1 = (ab)'(cd')' = (a'+b')(c'+d) = a'c' + a'd + b'c' + b'd$

$T_2 = (a'b'c)'d' = (a+b+c')d' = ad' + bd' + c'd'$

$T_3 = a'bc$

$T_4 = (ab')'(cd)' = (a'+b)(c'+d') = a'c' + a'd' + bc' + bd'$

$T_5 = b'c'd$

Combining all four terms:

$$F' = a'c' + a'd + b'c' + b'd + ad' + bd' + c'd' + a'c' + a'd' + bc' + bd' + a'bc + b'c'd$$

$$F' = a'c' + a'(d+d') + b'c'(1+d) + b'd + ad' + c'd' + bc' + a'bc + bd'$$

$$F' = a'c' + a' + b'c' + b'd + ad' + c'd' + bc' + a'bc + bd'$$

$$F' = a' + c'(b'+b) + b'd + ad' + c'd' + a'bc$$

$$F' = \underline{a'} + \underline{c'} + b'd + ad' + \underline{c'd'} + \underline{a'bc}$$

$$F' = a' + c' + b'd + ad'$$

$$F' = (a'+a)(a'+d') + c' + b'd \qquad \because x+yz = (x+y)(x+z)$$

$$F' = a' + d' + c' + b'd$$

$$F' = a' + c' + (d'+b')(d'+d)$$

$$\boxed{F' = a' + b' + c' + d'}$$

Question: 06

(a)

Product of Sum (POS Form)

(b) $(A'+B)(C'D)^2(B'+C+D))$

$(A'+B)(B'C'D + 0 + C'D)$

$\rightarrow (A'+B)(C'D)$

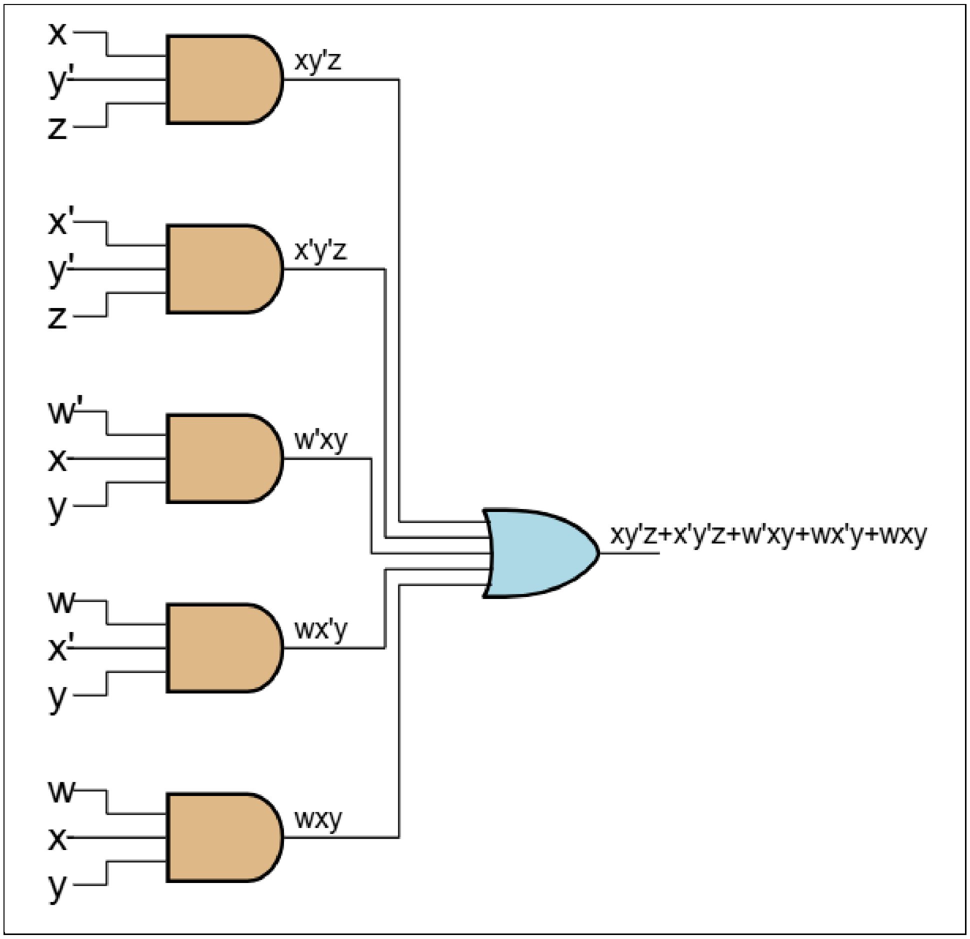
$\rightarrow A'C'D + BC'D$

Question 7

(a) 2pts

w	x	y	z	xy'z	x'y'z	w'xy	wx'y	wxy	F
0	0	0	0	0	0	0	0	0	**0**
0	0	0	1	0	**1**	0	0	0	**1**
0	0	1	0	0	0	0	0	0	**0**
0	0	1	1	0	0	0	0	0	**0**
0	1	0	0	0	0	0	0	0	**0**
0	1	0	1	**1**	0	0	0	0	**1**
0	1	1	0	0	0	**1**	0	0	**1**
0	1	1	1	0	0	**1**	0	0	**1**
1	0	0	0	0	0	0	0	0	**0**
1	0	0	1	0	**1**	0	0	0	**1**
1	0	1	0	0	0	0	**1**	0	**1**
1	0	1	1	0	0	0	**1**	0	**1**
1	1	0	0	0	0	0	0	0	**0**
1	1	0	1	**1**	0	0	0	0	**1**
1	1	1	0	0	0	0	0	**1**	**1**
1	1	1	1	0	0	0	0	**1**	**1**

(b) **2pts**



Question: 07

(c) 2pts

$y'z(x+x') + xy(w'+w) + wx'y$

$y'z + xy + wx'y$

$y'z + y(x+wx') = y'z + y(x+w)$

$\boxed{y'z + yx + yw}$

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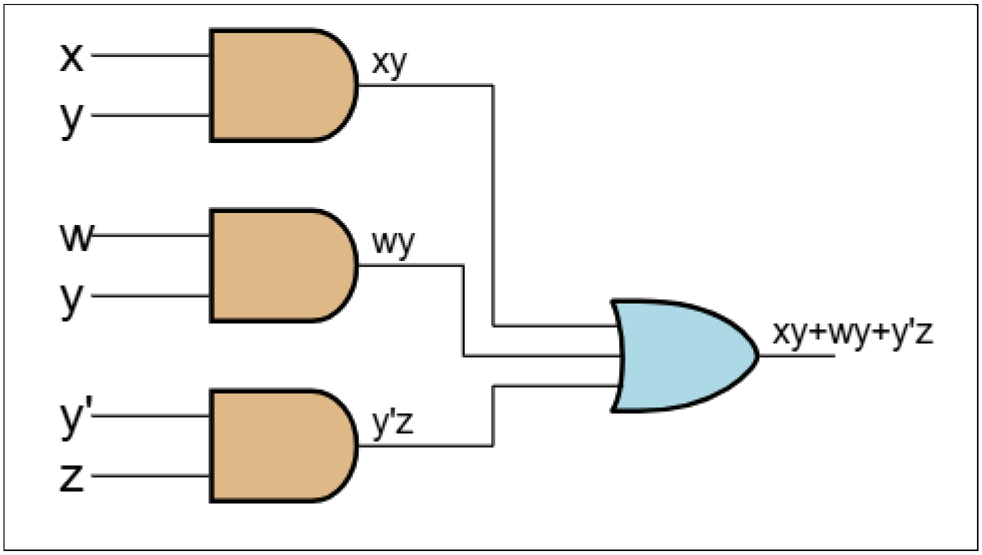
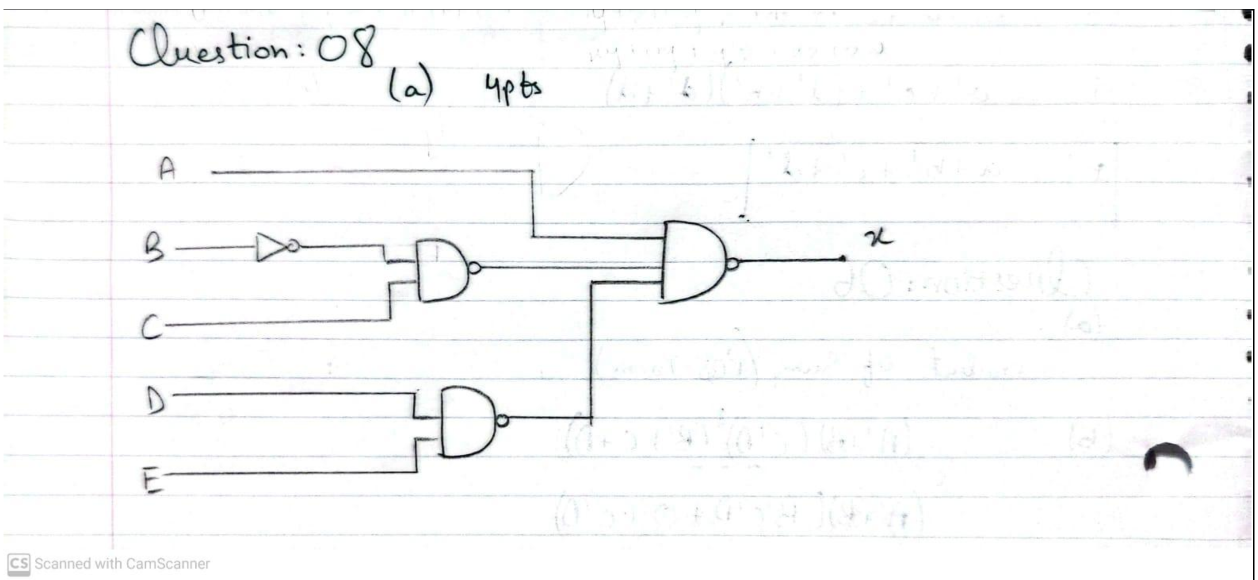
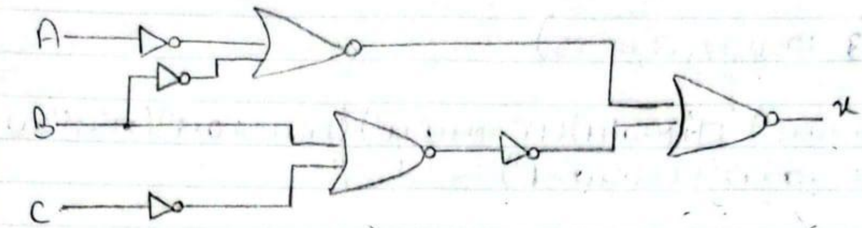
(d) **2pts**



The circuit uses 4 gates, compared to 6 gates from the original expressions circuit



(b) 2pts



Question: 09

(a) 3pts

$$= (AB)'(C'D')'$$
$$\Rightarrow (A'+B')(C+D)$$

(b) 3pts

$$= (W'+X)(Y'XZ' + WZY)$$
$$= W'XY'Z' + XY'Z' + WZXY$$
$$= XY'Z' + WXYZ$$

(c) 3pts

$$= (P'+Q)(RS'PQ + RS')$$
$$= P'S'R + RS'PQ + RS'Q$$
$$= P'RS' + RS'Q$$

(d) 3pts

lets first find the minterms

$$F' = \overset{7}{\overset{0111}{M'NOP}} + \overset{5}{\overset{0101}{M'NO'P}} + \overset{6}{\overset{0110}{M'NOP'}} + \overset{4}{\overset{0100}{M'NO'P'}} + \overset{9}{\overset{1001}{MN'O'P}} + \overset{1}{\overset{0001}{M'N'O'P}} +$$
$$\underset{8}{\underset{1000}{MN'O'P'}} + \underset{0}{\underset{0000}{M'N'O'P'}} + \underset{2}{\underset{0010}{M'N'OP'}}$$

$$F' = \sum(0, 1, 2, 4, 5, 6, 7, 8, 9)$$

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For maxterms

$$F' = \Pi(3, 10, 11, 12, 13, 14, 15)$$

$$F' = (M+N'+O'+P')(M'+N+O'+P)(M'+N+O'+P')(M'+N'+O+P)(M'+N'+O+P')$$
$$(M'+N'+O'+P)(M'+N'+O'+P')$$

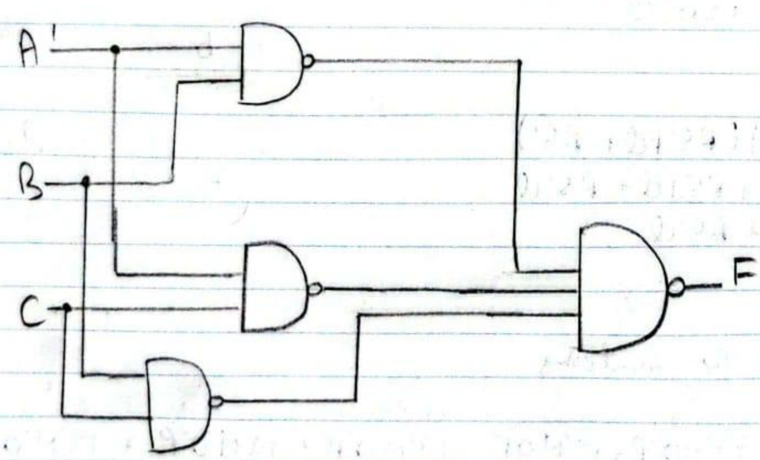
(e) 4pts

$$F = (A'+B'+C)(A'B+A'C+B+BC)$$

$$= A'B + A'C + A'BC + A'B'C + A'\overline{BC} + A'C + BC$$

$$= A'C(1+B+B') + A'B + BC$$

$$F = A'B + A'C + BC$$



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Question : 10

(a) 3pts

WX \ YZ	00	01	11	10
00	0			0
01		0		
11		0		
10	0	0	0	

$F = (W+X+Z)(W'+X+Y)(W'+X+Z')(X'+Y+Z')(W'+Y+Z')$

(b) 3pts

AB \ CD	00	01	11	10
00	1	X		1
01		X		1
11			1	1
10		X		1

$F = A'B'C' + ABC + CD'$

Question: 11

(a) 5pts

AB \ CD	00	01	11	10
00		1		
01	X	1		1
11	1	1		1
10			1	

$X = A'C'D + BC' + BCD' + AB'CD$

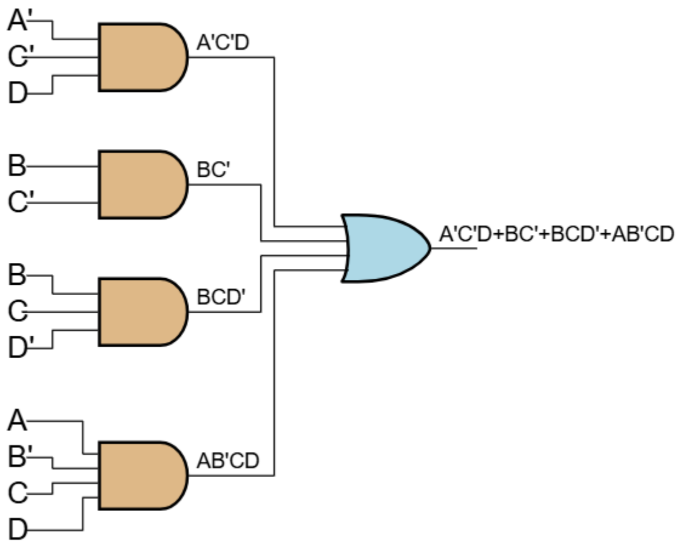
(b)

AB \ CD	00	01	11	10
00	0		0	
01	X	0	0	
11			0	
10		0	0	0

$Y = (A+C+D)(A+B'+D')(C'+D')(A'+B+D')(A'+B+C')$

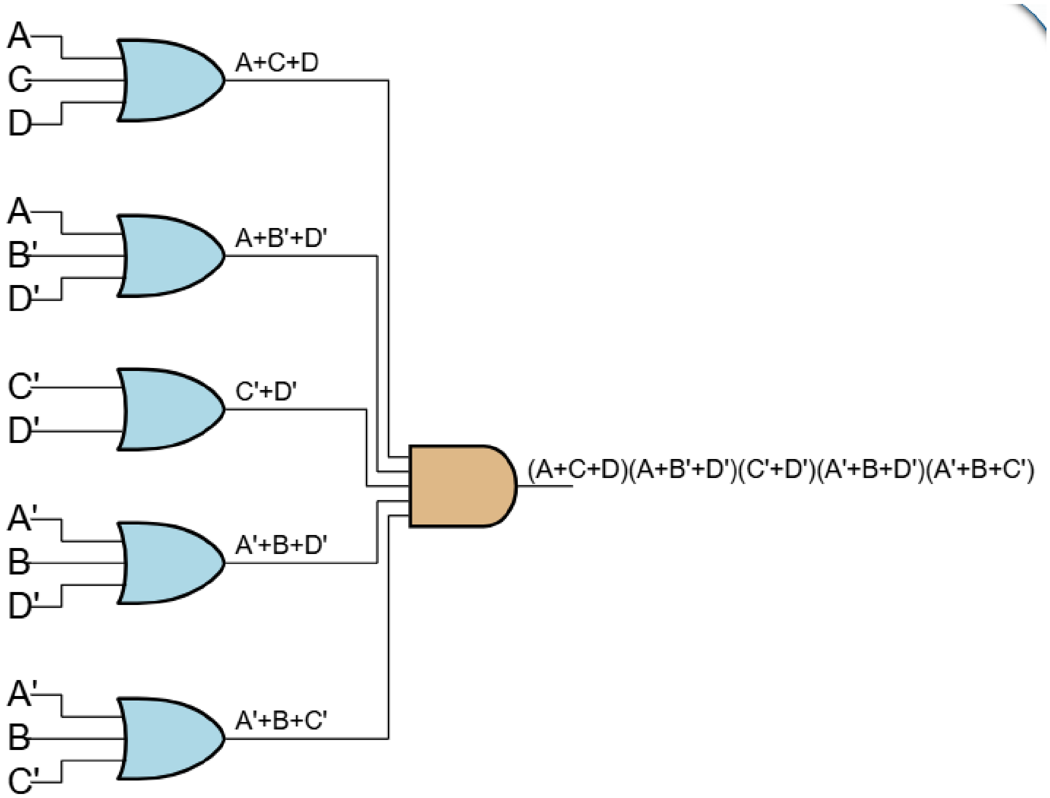
(b) **5pts**

X:



A'
C'
D
A'C'D
B
C'
BC'
B
C
D'
BCD'
A
B'
C
D
AB'CD
A'C'D+BC'+BCD'+AB'CD

Y:



A
C
D
A+C+D
A
B'
D'
A+B'+D'
C'
D'
C'+D'
A'
B
D'
A'+B+D'
A'
B
C'
A'+B+C'
(A+C+D)(A+B'+D')(C'+D')(A'+B+D')(A'+B+C')

Question: 12

(a) 3pts

$(W+Y+[(W+Y)Z]')(WY')$

$(W+Y+((W+Y)'+Z'))(WY')$

$(W+Y+W'Y'+Z')(WY')$

$WY'+WY'Z = WY'$

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W	X	Y	Z	Y’	F = W·Y’
0	0	0	0	1	0
0	0	0	1	1	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	1	1
1	1	0	1	1	1

W X Y Z Y' F = W·Y'

1 1 1 0 0 0

1 1 1 1 0 0

(C)

Karnaugh map

	$\overline{y}\overline{z}$	$\overline{y}z$	yz	$y\overline{z}$
$\overline{w}\overline{x}$	0	0	0	0
$\overline{w}x$	0	0	0	0
wx	1	1	0	0
$w\overline{x}$	1	1	0	0

(D)

Only 1 gate will be used as compared to the 5 gates in the original logic

Question: 13 (a) 4pts

WX \ YZ	00	01	11	10
00	X	1	X	1
01		X	1	1
11	X	1	1	1
10	1			

$F = WY'Z' + W'X' + WX + XY$

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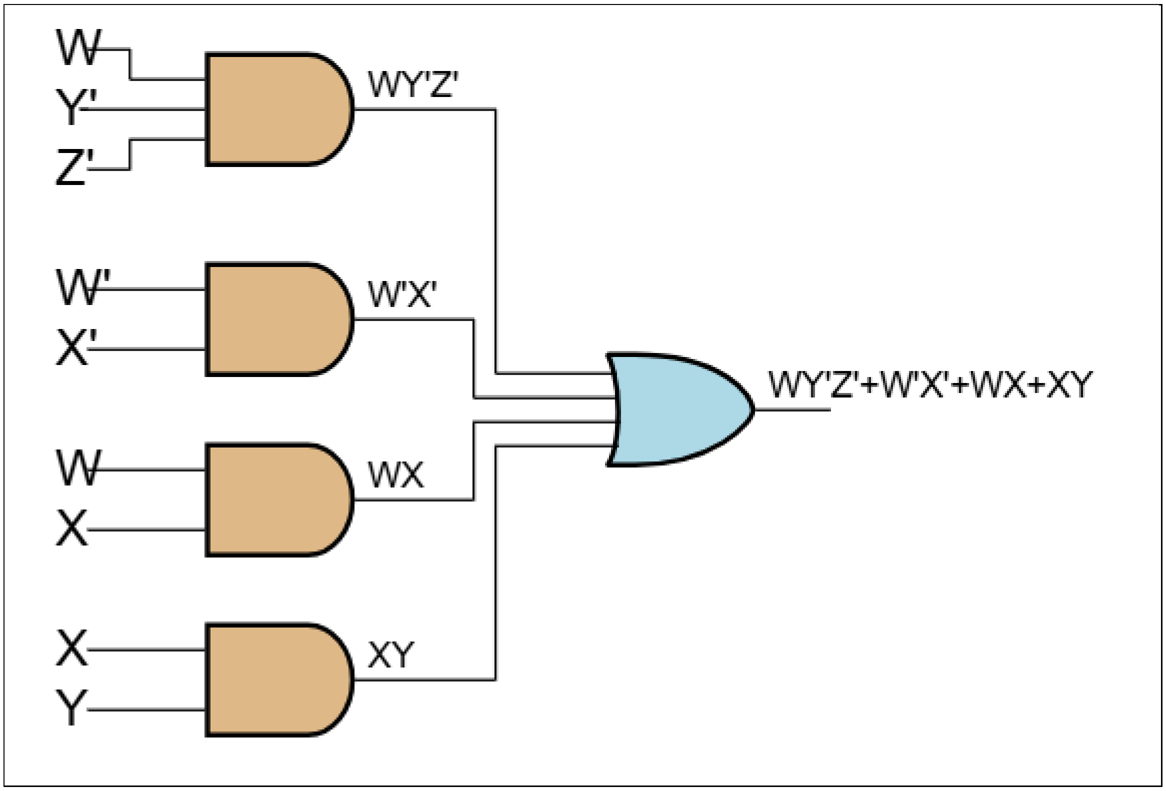
(b)



(c) 3pts

There are 8 minterms in the original expression and each will have 4 literals.
Whereas in the simplified form, there are 4 terms with 2 and 3 literals.

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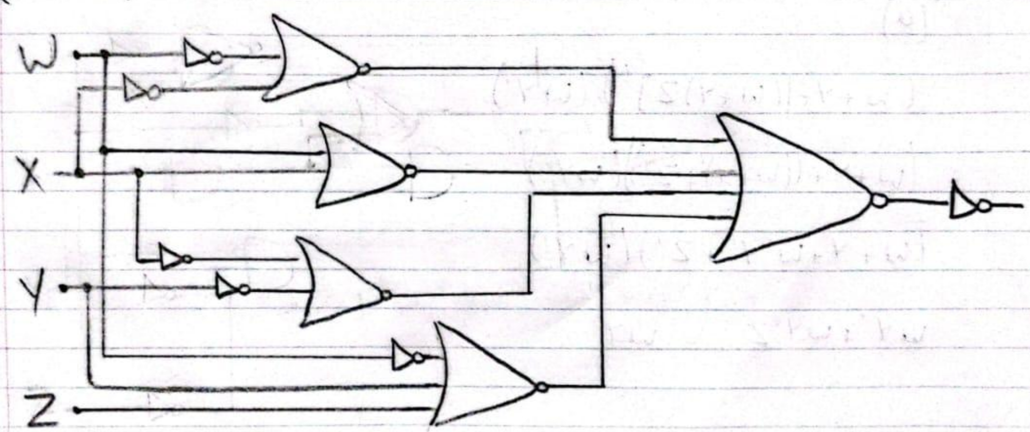
(d) 3pts



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